

Dossier Space Pilot — Session 10 Work Log (Continued)

Session ID: DS-20260811-S11 (prep work, still in Session 10 closure)

Design Lead: Claude

Operator: Ken Tombs

Date: August 11, 2026

Focus: v1.4 Custody Server build, startup validation, smoke test preparation

Work Completed

1. v1.4 Fresh Build (Clean Version Separation)

Action: Rebuilt MCP Custody Server as v1.4 (not patching v1.3)

Rationale: v1.3 was built without JSON-RPC protocol layer. Rather than patch in place (risk of incomplete repairs re-emerging), a clean v1.4 was created with fresh SESSION_ID.

Deliverable: `dossier_custody_server_v1_4.py` (34,981 bytes)

2. FastMCP 1.29.0 API Discovery and Correction

Issue Encountered: Multiple attempts to run v1.4 failed because the JSON-RPC protocol binding pattern was wrong for FastMCP 1.29.0.

Debugging Process:

1. First attempt: Used `async with server:` — failed (Server object doesn't support async context manager)
2. Second attempt: Used `async with stdio_server.stdio_server(server) as (read_stream, write_stream):` — failed (stdio_server is a function, not a module)
3. **Diagnostic workstation testing** (Ken executing):

```
python -c "from mcp.server import Server; s = Server('test'); print([m for m in dir(s) if not m.startswith('_')])"
```

→ Revealed available methods: `['call_tool', 'completion', 'create_initialization_options', ..., 'run', ...]`

1. **Further testing:**

```
python -c "from mcp import stdio_server; import inspect; print(inspect.signature(stdio_server))"
```

→ Showed: `(stdin: anyio.AsyncFile[str] | None = None, stdout: anyio.AsyncFile[str] | None = None)`

1. Final testing:

```
python -c "from mcp import stdio_server; s = stdio_server(); print(type(s))"
```

→ Confirmed: `<class 'contextlib._AsyncGeneratorContextManager'>`

Correct Pattern Identified:

```
async with stdio_server() as (read_stream, write_stream):
    await server.run(
        read_stream,
        write_stream,
        server.create_initialization_options()
    )
```

Lesson: API discovery through diagnostic testing on the actual workstation environment was essential. Assuming patterns from documentation alone led to multiple failures. Ken's willingness to validate each step prevented wasted iterations.

3. Startup Validation (Successful)

Test Executed:

```
python dossier_custody_server_v1_4.py
```

Output (First 20 lines):

```
2026-08-11 15:00:36,496 [INFO] Starting Dossier Space MCP Custody Server v1.4
2026-08-11 15:00:36,496 [INFO] Session: DS-20260811-S11-v1.4
2026-08-11 15:00:36,497 [INFO] Ledger: E:\AAll XPlain\Dossier Space Pilot\Governance
Ledger\session-ledger.jsonl
2026-08-11 15:00:36,497 [INFO] Corpus Root: E:\AAll XPlain\Dossier Space Pilot\ENRON
Data Set Test 1\enron_mail_20150507\maildir
2026-08-11 15:00:36,497 [INFO] Mode: Persistent MCP Service with JSON-RPC Protocol (v
1.4 clean rebuild)
2026-08-11 15:00:36,504 [INFO] Action counter recovered: highest action was A03, cont
inuing from A04
2026-08-11 15:00:36,505 [INFO] DS-20260811-S11-v1.4-A04: Ledger entry emitted (primar
y)
2026-08-11 15:00:36,505 [INFO] Session DS-20260811-S11-v1.4 started
2026-08-11 15:00:36,505 [INFO] CustodyServer initialized: corpus_root=E:\AAll XPlain
\Dossier Space Pilot\ENRON Data Set Test 1\enron_mail_20150507\maildir, task_id=DS-20
260811-S11-T01
2026-08-11 15:00:36,505 [INFO] MCP Custody Server v1.4 initialized and ready
2026-08-11 15:00:36,505 [INFO] Available tools:
2026-08-11 15:00:36,505 [INFO] - search_messages(custodian, keyword, date_contains)
2026-08-11 15:00:36,505 [INFO] - list_folder(custodian)
2026-08-11 15:00:36,505 [INFO] - read_file(filepath)
2026-08-11 15:00:36,505 [INFO] - get_headers(filepath)
2026-08-11 15:00:36,505 [INFO] - hash_document(filepath)
2026-08-11 15:00:36,505 [INFO] - verify_hash(filepath, expected_hash)
2026-08-11 15:00:36,506 [INFO] Server ready for MCP client connections (JSON-RPC prot
ocol active)
2026-08-11 15:00:36,506 [INFO] Starting MCP server on stdio...
2026-08-11 15:00:36,511 [INFO] MCP server listening on stdio
```

Validation Checklist:

- ✓ Session ID correctly formatted: `DS-20260811-S11-v1.4`
- ✓ Ledger initialized: primary write successful
- ✓ Action counter recovered: A03 → A04 (continuity from prior runs)
- ✓ CustodyServer initialized with correct corpus path
- ✓ All six tools registered and listed
- ✓ Server bound to stdio (JSON-RPC protocol active)
- ✓ Persistent (not exiting after initialization)

Status: ✓ STARTUP VALIDATED

Next Steps: Smoke Test

Objective: Validate end-to-end tool execution through MCP protocol

Test Plan:

1. Establish MCP client connection to running server
2. Execute `search_messages(custodian="farmer-d", keyword="deal")`
3. Verify:
 - Tool call routed correctly to CustodyServer
 - Action ID generated and sequenced
 - Ledger entry created with full payload
 - Result set persisted and hashed
 - JSON-RPC response returned with results

Expected Outcome: Similar to v1.2 smoke test (Session 9.5) — 1,147 candidates found for "deal" in farmer-d, ledger entries validated, result file checksummed.

Governance Notes

Workstation Configuration Validated:

- Python 3.14.6 ✓
- pip 26.1.2 ✓
- FastMCP 1.29.0 ✓
- Corpus path accessible ✓
- Governance Ledger directory accessible ✓

Session Status: Still in **Session 10 closure prep**. Smoke test will determine readiness for Session 11 execution sessions.

Log prepared by: Claude (Design Lead)

Validated by: Ken Tombs (Operator)

Date logged: August 11, 2026, 15:00 UTC